

From topology to combinatorics via computability theory

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Motivation

Combinatorics played an important role in the history and development of reverse mathematics.

This tutorial will attempt to survey a few of the central ideas, methods, and themes of this development.

We will center our discussion around the following paper:

- ▶ Heidi Benham, Andrew DeLapo, Damir D. Dzhamalov, Reed Solomon, and Java Darleen Villano, "The Ginsburg–Sands theorem and computability theory", *Advances in Mathematics*, vol. 444, 109618 (2024).

This analyzes a theorem in topology that turns out to have surprising connections to Ramsey's theorem and similar combinatorial statements.

As such, it serves as a conduit for going [from topology to combinatorics](#).

Lecture 1

The Ginsburg-Sands theorem

Theorem (Ginsburg & Sands, 1979)

Every infinite topological space has a subspace homeomorphic to ω with one of the following topologies:

- ▶ indiscrete topology; (only $\emptyset$ and ω .)
- ▶ initial segment topology; ($\emptyset$, ω , and $[0, n]$ for all $n \in \omega$.)
- ▶ final segment topology; ($\emptyset$, ω , and $[n, \infty]$ for all $n \in \omega$.)
- ▶ discrete topology; (all subsets of ω .)
- ▶ cofinite topology; ($\emptyset$, ω , and all cofinite subsets of ω .)

A homogeneity property

One form of the **pigeonhole principle** states that given $c : \omega \rightarrow \{R, B\}$ there is an infinite $H \subseteq \omega$ such that $c(x) = R$ for all $x \in H$ or $c(x) = B$ for all $x \in H$.

(H is said to be a **homogeneous** set for c with color R or B , as the case may be)

Pigeonhole principle

- ▶ A set can be homogeneous with at most one of R or B .
- ▶ Every subset of a homogeneous set is homogeneous with the same color.

G-S theorem

- ▶ A space can be homeomorphic to at most one of the above topologies.
- ▶ Every subspace of such a space is also homeomorphic to that same topology.

Original proof of G–S theorem

Let X be an infinite topological space. WOLOG, assume X is countable.

Define an **equivalence relation** $\sim$ on X by $x \sim y$ if $x \in \text{cl}(y)$ and $y \in \text{cl}(x)$.

($x \in \text{cl}(y)$ if every open set containing x also contains y .)

- If $[x]_{\sim}$ is infinite for some x , let $Y = [x]_{\sim}$. Then Y is homeomorphic to ω with the indiscrete topology.

So we next consider the case that $[x]_{\sim}$ is finite for all $x \in X$.

- By choosing representatives, we can assume X is T_0 .

Define a **partial order** $\trianglelefteq$ on X by $x \trianglelefteq y$ if $x \in \text{cl}(y)$.

(So necessarily, $y \notin \text{cl}(x)$.)

Original proof of G–S theorem

We have $x \trianglelefteq y$ if $x \in \text{cl}(y)$.

Every infinite partial order contains an **infinite chain** or an **infinite antichain**.

Suppose X contains an infinite $\trianglelefteq$ -chain. (So, a $\trianglelefteq$ -linearly ordered subset.)

Every infinite linear order contains either:

- ▶ an infinite ascending sequence; (Order type ω .)
- ▶ or an infinite descending sequence. (Order type ω^* .)

A $\trianglelefteq$ -ascending sequence is homeomorphic to the final segment topology.

A $\trianglelefteq$ -descending sequence is homeomorphic to the initial segment topology.

Original proof of G–S theorem

We have $x \trianglelefteq y$ if $x \in \text{cl}(y)$. We now assume X has no infinite $\trianglelefteq$ -chain. WOLOG, we can also assume that X itself is T_1 .

(For all $x \neq y$ in X , there exist open U, V with $x \in U \setminus V$ and $y \in V \setminus U$.)

Our goal is to show that X has an infinite discrete subspace or one homeomorphic to the cofinite topology. Suppose it has none of the latter kind.

- ▶ Since X is T_1 , points are closed in X . Hence every cofinite subset of X is open.
- ▶ Our assumption implies that for every infinite $Y \subseteq X$ there is an open set U in X such that $U \cap Y \neq \emptyset$ and $Y \setminus U$ is infinite.

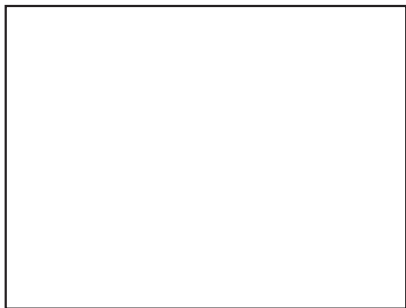
We now construct a discrete subspace $Y = \{x_0, x_1, \dots\}$ of X inductively.

- ▶ We define Y along with $U_0, U_1, \dots$ open in X such that $x_i \in U_j$ iff $i = j$.

Original proof of G–S theorem

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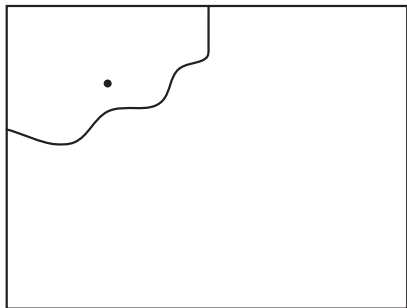
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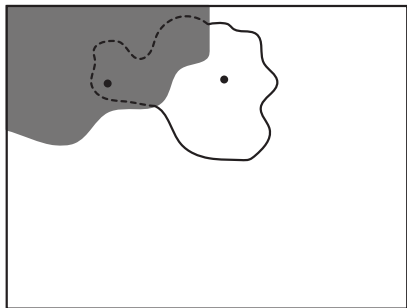


- There is a U so that $U \neq \emptyset$ and $X \setminus U$ is infinite.
- Choose $x_0 \in U$ and let $U_0 = U$.

Original proof of G-S theorem

We now construct a discrete subspace $Y = \{x_0, x_1, \dots\}$ of X inductively.

► We define Y along with $U_0, U_1, \dots$ open in X such that $x_i \in U_j$ iff $i = j$.

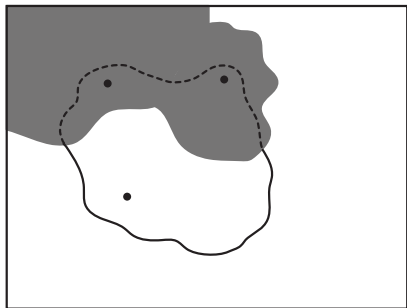


- There is a U so that $U \cap (X \setminus U_0) \neq \emptyset$ and $(X \setminus U_0) \setminus U$ is infinite.
- Choose $x_1 \in (X \setminus U_0) \setminus U$ and let $U_1 = U \setminus \{x_0\}$.

Original proof of G–S theorem

We now construct a discrete subspace $Y = \{x_0, x_1, \dots\}$ of X inductively.

► We define Y along with $U_0, U_1, \dots$ open in X such that $x_i \in U_j$ iff $i = j$.



- There is a U so that $U \cap (X \setminus (U_0 \cup U_1)) \neq \emptyset$ and $(X \setminus (U_0 \cup U_1)) \setminus U$ is infinite.
- Choose $x_2 \in (X \setminus (U_0 \cup U_1)) \setminus U$ and let $U_2 = U \setminus \{x_0, x_1\}$.

Recapping

The original proof of the G–S theorem naturally breaks into three components:

- ▶ Construction of the $\sim$ and $\trianglelefteq$ relations using the closure operator.
 - ▶ Application of combinatorial facts about partial and linear orders.
 - ▶ An inductive construction to handle the T_1 case.
-

What can computability theory and reverse math try to answer?

- ▶ What axioms are needed to prove this theorem?
- ▶ Is the use of combinatorics that we saw necessary?
- ▶ Is this the optimal proof in any sense?
- ▶ Can the breakdown above be made precise in any sense?

Language of second-order arithmetic

L_2 is a **two-sorted** (but still first-order) language of signature

$$\{0, S, +, \cdot, <, =, \in\}.$$

- ▶ first-sort variables $x, y, z, \dots$; (Intended to range over numbers.)
- ▶ second-sort variables $X, Y, Z, \dots$; (Intended to range over sets of numbers.)
- ▶ $\in$ only relates a first-sort term to a second-sort term;
- ▶ $=$ only holds between first-sort terms.

($X = Y$ abbreviates $(\forall x)[x \in X \leftrightarrow x \in Y]$.)

For $n \geq 0$, we define the syntactic classes Σ_n^0 , Π_n^0 , Σ_n^1 , and Π_n^1 in L_2 as usual.

L_2 formulas may involve **parameters** from models. (More on this later.)

Full second-order arithmetic

Z_2 is the L_2 theory consisting of the following axioms and schemata:

- ▶ the axioms of PA^- , e.g., (Discrete ordered commutative semiring axioms.)
 $(\forall x)[x + 0 = x], \quad (\forall x, y)[x + S(y) = S(x + y)],$
 $(\forall x, y, z)[(x > 0 \wedge y < z) \rightarrow x \cdot y < x \cdot z], \quad \neg(\exists x)[S(x) = 0], \quad \dots$
- ▶ for each formula $\varphi(x)$ with only x free (and possibly set parameters),
the axiom $(\exists X)(\forall x)[x \in X \leftrightarrow \varphi(x)];$ (Comprehension.)
- ▶ for each formula $\varphi(x)$ with only x free (and possibly set parameters),
the axiom $(\varphi(0) \wedge (\forall x)[\varphi(x) \rightarrow \varphi(S(x))]) \rightarrow (\forall x)\varphi(x).$ (Induction.)

Fragments of Z_2 give rise to **subsystems of second-order arithmetic**.

Recursive comprehension

RCA_0 consists of PA^- together with:

- ▶ induction restricted to Σ_1^0 formulas; ($\text{I}\Sigma_1^0$)
- ▶ comprehension restricted to Σ_1^0 formulas $\varphi(x)$ such that ($\Delta_1^0\text{-CA}$)
 $(\forall x)[\varphi(x) \leftrightarrow \psi(x)]$ for some Π_1^0 formula $\psi(x)$.

RCA_0 proves primitive recursion and minimization.

- ▶ Allows for sets to be constructed recursively.
- ▶ Care must be taken in the case of some constructions (e.g., finite injury arguments) that totality/correctness can be verified using $\text{I}\Sigma_1^0$.

RCA_0 proves that the pairing function $\langle \cdot, \cdot \rangle$ is one-to-one, and so can do basic codings of pairs, finite sequences of numbers, functions $\mathbb{N} \rightarrow \mathbb{N}$, etc.

(Working in RCA_0 , we use $\mathbb{N}$ to denote the set of first-sort terms.)

Arithmetical comprehension

ACA_0 is RCA_0 + comprehension restricted to Σ_1^0 formulas. (Σ_1^0 -CA)

Actually, ACA_0 proves comprehension for all arithmetical formulas. (Σ_n^0 -CA)

- ▶ E.g., fix $\varphi(x) \equiv (\exists y)\psi(x, y)$ where ψ is Π_1^0 . Then ACA_0 proves $X = \{\langle x, y \rangle : \neg\psi(x, y)\}$ exists, and using it as a parameter, also that $\{x : (\exists y)\langle x, y \rangle \notin X\}$ exists.

ACA_0 also proves induction for all arithmetical formulas. ($I\Sigma_n^0$)

Over RCA_0 , $\text{ACA}_0 \leftrightarrow (\forall f: \mathbb{N} \rightarrow \mathbb{N})[\text{ran}(f) \text{ exists}]$.

- ▶ Arguing in RCA_0 , fix $\varphi(x) \equiv (\exists y)\psi(x, y)$ where ψ is Σ_0^0 . Say $\psi(x_0)$ holds. Define $f: \mathbb{N} \rightarrow \mathbb{N}$ by $f(\langle x, y \rangle) = x$ if $\psi(x, y)$ and $f(\langle x, y \rangle) = x_0$ otherwise. Then $\text{ran}(f) = \{x : \varphi(x)\}$.

Models and ideals

An **L₂-structure** is a tuple $\mathcal{M} = \{M, \mathcal{S}^{\mathcal{M}}, 0^{\mathcal{M}}, S^{\mathcal{M}}, +^{\mathcal{M}}, \cdot^{\mathcal{M}}, <^{\mathcal{M}}\}$ such that:

- ▶ $\mathcal{S}^{\mathcal{M}} \subseteq \mathcal{P}(M)$;
- ▶ $=$ and $\in$ are interpreted as equality in M and set membership, resp.

If $\mathcal{M} \models Z_2$ then $\{M, 0^{\mathcal{M}}, S^{\mathcal{M}}, +^{\mathcal{M}}, \cdot^{\mathcal{M}}\} \models \text{PA}$. But it may be nonstandard.

An **ω -model** is an L₂-structure $\mathcal{M}$ in which $M = \omega$, all symbols are interpreted as the obvious arithmetical operations, and $\mathcal{S}^{\mathcal{M}}$ is a Turing ideal.

($\mathcal{S}^{\mathcal{M}}$ is closed under $\leq_T$ and $\oplus$.)

- ▶ Every ω -model satisfies RCA_0 .
- ▶ The ω -models satisfying ACA_0 are precisely the jump ideals.

($\mathcal{S}^{\mathcal{M}}$ is closed under $X \mapsto X'$.)

CSC spaces

The following definition is made in RCA_0 .

Definition (Dorais). A **CSC space** is a tuple $\langle X, \mathcal{U}, k \rangle$ where:

- ▶ $X \subseteq \mathbb{N}$ is an infinite set;
- ▶ $\mathcal{U} = \langle U_n : n \in \mathbb{N} \rangle$ is a sequence of subsets of X containing X and $\emptyset$;
- ▶ $k : X \times \mathbb{N}^2 \rightarrow \mathbb{N}$ is such that if $x \in U_n \cap U_m$ then $x \in U_{k(x,n,m)} \subseteq U_n \cap U_m$.

Open sets are given in terms of codes: $O \subseteq \mathbb{N}$ codes $\bigcup_{n \in \mathbb{N}} U_n$. (Σ_1^0 in X .)

A **subspace** of a X is a CSC space of the form

$$\langle Y, \langle U_n \cap Y : n \in \mathbb{N} \rangle, k \upharpoonright Y \times \mathbb{N}^2 \rangle$$

for some infinite $Y \subseteq X$.

Formalizing subspace properties

Let $\langle \mathbb{N}, \mathcal{U}, k \rangle$ be a CSC space, with $\mathcal{U} = \langle U_n : n \in \mathbb{N} \rangle$.

- ▶ X is **indiscrete** if $(\forall n)[U_n = \emptyset \vee U_n = X]$.
- ▶ X is **discrete** if $(\forall x)(\exists n)[U_n = \{x\}]$.
- ▶ X has the **cofinite topology** if $(\forall n)[U_n = \emptyset \vee U_n \subseteq^* X]$.
- ▶ X has the **initial segment topology** if there exists a bijection $h : \mathbb{N} \rightarrow X$ such that $(\forall n)(\exists j)[U_n = \{h(i) : i \leq j\} \vee U_n = X]$.
- ▶ X has the **final segment topology** if there exists a bijection $h : \mathbb{N} \rightarrow X$ such that $(\forall n)(\exists j)[U_n = \{h(i) : i \geq j\} \vee U_n = \emptyset]$.

One can also give an arithmetical definition of the initial/final segment topologies. Up to passing to a subspace, these are equivalent over RCA_0 .

The Ginsburg–Sands theorem, restated

The following statement is defined in RCA_0 .

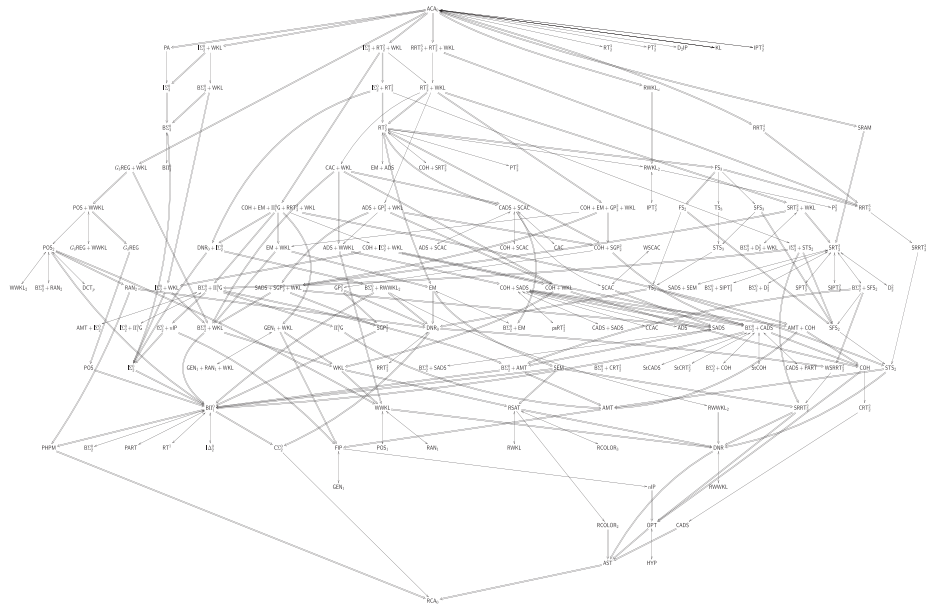
GS. Every infinite CSC space has an infinite subspace that is indiscrete, has the initial segment topology, has the final segment topology, is discrete, or has the cofinite topology.

Like many theorems, GS has the form $(\forall X)[\varphi(X) \rightarrow (\exists Y)\psi(X, Y)]$, where φ and ψ are properties of the sets X and Y (often arithmetical).

We often refer to such a theorem P as an **(instance-solution) problem**.

- ▶ the **P-instances** of the problem are all X such that $\varphi(X)$;
- ▶ the **P-solutions** to X are all Y such that $\psi(X, Y)$.

The reverse mathematics zoo



The most important theorem of combinatorics

Definition. Fix $X \subseteq \omega$ and $n \geq 1$.

- ▶ Let $[X]^n = \{F \subseteq X : |F| = n\}$.
- ▶ For $k \geq 1$, a **k -coloring of $[X]^n$** is a map $c : [X]^n \rightarrow k$. ($k = \{0, \dots, k-1\}$.)
- ▶ $H \subseteq X$ is **homogeneous for c with color $i < k$** if $c(F) = i$ for all $F \in [H]^n$.

Ramsey's theorem.

For every $n, k \geq 1$ and every $c : [\omega]^n \rightarrow k$, there exists an infinite homogeneous set for c .

In L_2 , we denote this statement by **RT**. For $n, k \geq 1$, **RT** ^{n} and **RT** _{k} ^{n} denote the restrictions to colorings of $[\mathbb{N}]^n$ and k -colorings of $[\mathbb{N}]^n$, respectively.

An upper bound

Theorem (Jockusch and Simpson). For all $n, k \geq 1$, ACA_0 proves RT_k^n .

The proof is by induction on n .

(The induction is external!)

The base case $n = 1$ is trivial. (Why?) So assume the result for some $n \geq 1$.

Arguing in ACA_0 , fix $c : [\mathbb{N}]^{n+1} \rightarrow k$.

Claim. There is an infinite set $X = \{x_0 < x_1 < \dots\}$ and a map $d : [X]^n \rightarrow k$ such that for every $F \in [X]^n$ and every $y > F$ in X we have $c(F, y) = d(F)$.

Apply RT_k^n to find an infinite homogeneous set H for d . Then H is also homogeneous for c .

(How? Formally, we can only apply RT_k^n to colorings of $[\mathbb{N}]^n$.)

An upper bound

Claim. There is an infinite set $X = \{x_0 < x_1 < \dots\}$ and a map $d : [X]^n \rightarrow k$ such that for every $F \in [X]^n$ and every $y > F$ in X we have $c(F, y) = d(F)$.

We prove this by recursion.

Let $\{x_0, \dots, x_{n-1}\}$ be the least n elements of $\mathbb{N}$. Let $d(x_0, \dots, x_{n-1})$ be the least i such that $\{y > x_{n-1} : c(x_0, \dots, x_{n-1}, y) = i\}$ is infinite.

Now say we have $X_m = \{x_0, \dots, x_m\}$ for some $m \geq n - 1$, and $d(F)$ for every $F \in [X_m]^n$, and that $Y_m = \{y > x_m : (\forall F \in [X_m]^n)[c(F, y) = d(F)]\}$ is infinite.

Let $x_{m+1} = \min Y_m$. Let $F_0, \dots, F_s$ list all members of $\{[x_0, \dots, x_{m+1}]^n\}$ containing x_{m+1} . For each $t \leq s$ in order, define $d(F_t)$ to be the least i so that $\{y \in Y_m : c(F_t, y) = i \wedge (\forall u \leq t)[c(F_u, y) = d(F_u)]\}$ is infinite.

(How do you do this formally?)

CAC and ADS

CAC: Every partial order $\leq_P$ on $\mathbb{N}$ has an infinite chain or antichain.

Let $(L, \leq_L)$ be a linear order. A set $X \subseteq L$ is:

- ▶ an **ascending sequence** for L if $\{y \in X : y \leq_L x\}$ is finite for each $x \in X$;
- ▶ a **descending sequence** for L if $\{y \in X : x \leq_L y\}$ is finite for each $x \in X$.

ADS: Every linear order $\leq_L$ on $\mathbb{N}$ has an infinite ascending sequence or descending sequence.

It is not hard to show that $RT_2^2 \rightarrow CAC \rightarrow ADS$ over RCA_0 . In particular, both CAC and ADS are provable in ACA_0 .